

DARKROOM DATAFORGE

Wits-merSETA Darkroom • Document Intelligence & Research Dataset Platform

1. What is Darkroom DataForge?

Darkroom DataForge is an internal desktop-and-web workstation designed for the **Wits-merSETA Darkroom** team. Its core job is simple: **convert large, messy, semi-structured PDF reports, gazetted occupational frameworks, and qualification tables into clean, validated, machine-readable datasets** (CSV, JSON, Excel, and SQL-ready databases).

Instead of acting like a fragile 'PDF-to-Excel converter', DataForge functions as a rigorous data-engineering workstation where **every single record is traceable back to its source PDF page and verified against South African labour-market standards**.

2. Why Can't We Just Use ChatGPT or Regular AI?

A common question is: *'Why build a dedicated platform when someone can just paste a PDF into ChatGPT, Claude, or Gemini?'*

Generic consumer AI tools are designed for **conversation and creative summarization**, which makes them dangerous for empirical research and official policy datasets. Here is why regular AI fails where DataForge succeeds:

Dimension	Generic Consumer AI (ChatGPT / Copilot)	Darkroom DataForge Platform
Data Truth & Fidelity	Summarises, guesses missing text, and hallucinates plausible-sounding numbers when tables get complex.	Source is Single Truth: Extractor is strictly constrained. It never invents data and defaults to 'Needs Review'.
Code Integrity (Leading Zeroes)	Treats codes as numbers: turns OFO code 021301 into 21301 or SAQA ID 0042 into 42, breaking mergers.	Format Preserved: Explicitly protects leading zeroes and code masks for SAQA, OFO, and NQF structures.
Multi-Page Tables	Drops rows across page boundaries, loses sub-headers, and truncates 50+ page documents silently.	Table-Aware Chunking: Slices tables with overlap, reconstructing split rows and continuations seamlessly.
Evidence & Provenance	No provenance. If an auditor asks where a number came from, you cannot prove it.	Full Provenance: Every cell links to exact PDF page (#page=N) and extraction model.
Data Privacy & Air-Gap	Uploads sensitive research data to external US public clouds.	100% On-Premise: Supports local air-gapped AI (Ollama) so sensitive research data never leaves your machine.
Discrepancy Reconciliation	Single model answer with no second opinion or agreement scoring.	Multi-Model Verification: Compares two models side-by-side and flags disagreements for human sign-off.

3. How Does Darkroom DataForge Work? (The 5 Steps)

- **Step 1: Document Inspection & Layout Mapping:** The system scans the PDF, determines whether it is digital text or scanned image, calculates text density, and identifies all tables and continuation groups spanning multiple pages.
- **Step 2: Table-Aware Slicing & Schema Inference:** Multi-page tables are sliced with contextual overlap so split rows are never cut in half. The system infers the dataset schema (column names, types, primary keys) and lets the researcher review it before running extraction.

- **Step 3: Constrained AI Extraction:** The AI processes each chunk under strict contractual rules: it must preserve leading zeroes, copy raw text faithfully, and record the exact page number for every record.
- **Step 4: Dual Validation (Rules + Multi-Model Check):** Records are checked against South African standards (OFO 6-digit codes, SAQA IDs, NQF levels 1–10). In High-Accuracy mode, a second independent AI model extracts the same pages. An agreement score is calculated and any disagreement is flagged as a conflict.
- **Step 5: Human-in-the-Loop Review & Research Sign-Off:** Researchers review flagged discrepancies and validation warnings. Clicking any item jumps directly to the source PDF page (#page=N). Once verified, the lead researcher formally signs off on the dataset.

4. The Standard Research Export Bundle

When an extraction job completes, DataForge does not just output a raw CSV. It packages a complete, publication-ready research bundle:

- **Dataset Files (.csv, .xlsx, .json):** Clean, validated, standardized tabular dataset ready for Stata, R, Python, or SQL databases.
- **Data Dictionary (data_dictionary.csv):** Official codebook specifying column names, original source headers, data types, nullability, and descriptions.
- **Provenance Record (provenance.csv):** Row-by-row audit trail linking every record to its document ID, page number, and extraction method.
- **Extraction Manifest (extraction_manifest.json):** Cryptographic SHA-256 hash, timestamps, model metadata, and parameters for scientific reproducibility.

Summary for Stakeholders:

*Generic AI is an assistant that guesses and chats; **Darkroom DataForge is a laboratory instrument** that extracts, cross-checks, validates, and proves every data point. For Wits–merSETA, this guarantees that public policy decisions, skills planning reports, and academic papers are grounded in 100% verified evidence.*